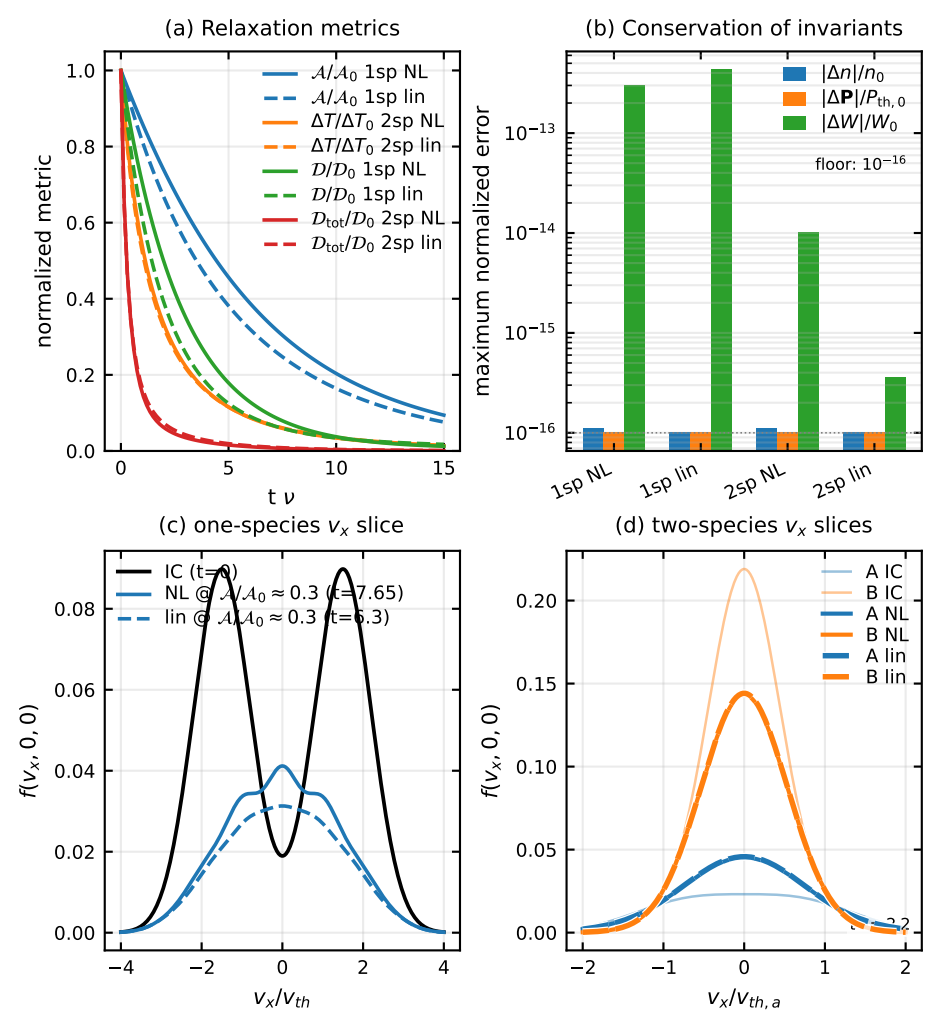
(a) Relaxation metrics
normalized metric
t ν
$\mathcal{A}/\mathcal{A}_0$ 1sp NL
$\mathcal{A}/\mathcal{A}_0$ 1sp lin
$\Delta T/\Delta T_0$ 2sp NL
$\Delta T/\Delta T_0$ 2sp lin
$\mathcal{D}/\mathcal{D}_0$ 1sp NL
$\mathcal{D}/\mathcal{D}_0$ 1sp lin
$\mathcal{D}_{\rm tot}/\mathcal{D}_0$ 2sp NL
$\mathcal{D}_{\rm tot}/\mathcal{D}_0$ 2sp lin
(b) Conservation of invariants
maximum normalized error
$|\Delta n|/n_0$
$|\Delta \mathbf{P}|/P_{\rm th,0}$
$|\Delta W|/W_0$
floor: 10^{-16}
1sp NL
1sp lin
2sp NL
2sp lin
(c) one-species v_x slice
$f(v_x, 0, 0)$
IC (t=0)
NL @ $\mathcal{A}/\mathcal{A}_0 \approx 0.3$ (t=7.65)
lin @ $\mathcal{A}/\mathcal{A}_0 \approx 0.3$ (t=6.3)
v_x/v_{th}
(d) two-species v_x slices
$f(v_x, 0, 0)$
A IC
B IC
A NL
B NL
A lin
B lin
$v_x/v_{th,a}$